

Do Small Vision-Language Models Route Language the Same Way in Pixels and in Tokens? A Mechanistic Investigation with Code-Switched Text

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Abstract

Vision-language models (VLMs) increasingly read text as pixels, including code-switched text such as a screenshot of a Hinglish tweet, yet it is unknown whether the attention heads that matter for a language when it arrives as text tokens are the same ones that matter when it arrives as a rendered image. We investigate this in four open VLMs under 3B parameters. We score attention heads with gradient-based head importance (LAHIS) on monolingual English, Spanish, and Hindi sentences presented in three conditions (as text, as a rendered image with a query, and as a rendered image alone), probe the vision encoders for script identity on rendered code-switched sentences, and ablate the top-scoring heads while the models read Hindi-English code-switched images. The answer depends on the model. In Qwen2-VL-2B and SmolVLM, text and image inputs recruit largely different heads (text-image Jaccard overlap 0.05–0.21, against 0.43–0.91 between the two image conditions), and in Qwen2-VL-2B the text heads sit almost entirely in the first two layers while the image heads do not. In InternVL3-2B the two sets nearly coincide, but only for Hindi (0.667). For PaliGemma-3B the overlap measure cannot distinguish either pattern from chance. All four vision encoders separate Latin from Devanagari script from their first layer. Zero-ablating the 20 top-scoring heads for either language destroys generation in every model (at most 1% of outputs stay fluent, against 71–76% when 20 random heads are ablated), so these heads are necessary for coherent output, but the ablation cannot show that they route language. A script-change metric counted this collapse as a change of script, which a fluency check exposes. We report these results as a per-model map rather than a general law, and release the pipeline, the corpus, and the per-head scores.

1 Introduction

A phone screenshot of a Hinglish tweet, a photographed handwritten note, and a chat screenshot pasted into a vision-language assistant all put the same problem to a VLM. It has to identify and route two languages within a single image, without the token-level segmentation a text-only language model gets for free.

A VLM has two main parts. A vision encoder turns an image into a sequence of tokens, much as a tokenizer turns text into a sequence of tokens. A language decoder then reads whichever tokens it is given, whether they came from text or from the vision encoder, and generates output text. The same decoder can therefore meet the same sentence along two different paths. Whether its attention heads handle the sentence the same way on both paths is not known.

Prior mechanistic work on multilingual routing has focused almost exclusively on the text-only case. [Liu et al. \(2026\)](#) introduce Language Attention Head Importance Scores (LAHIS) and show that a small set of attention heads in multilingual LLMs matters far more than the rest for language identification, discoverable via a single forward-backward pass with a differentiable head mask. [Nie et al. \(2025\)](#) trace language-confusion failures in English-centric LLMs to specific “confusion point” positions and final-layer transitions. [Wendler et al. \(2024\)](#) show that intermediate Llama-2 representations are closer to an English pivot than to the

input or output language. On the vision-language side, [Ye et al. \(2025\)](#) identify cross-lingual attention misalignment in VLMs and intervene on it to reduce multilingual object hallucination, and [Baek et al. \(2025\)](#) characterize “OCR heads” responsible for reading text embedded in images, but restrict their study to English text on English-centric models.

The question matters for how far existing interpretability work already takes us. If a VLM routes a sentence through the same heads whether it arrives as text or as pixels, findings about language heads from text-only work should carry over to the visual case. If the two paths use different heads, the visual path needs to be studied on its own. Motivated by this gap, we investigate two research questions and one exploratory question:

- **RQ1: Do text and image inputs share language-routing heads?** When the same sentence reaches a VLM as text tokens and as a rendered image, are the most important attention heads the same?
- **RQ2: Are the identified heads causally involved?** When a VLM reads code-switched text from an image, does ablating the heads that gradient attribution ranks highest change the language of its output, beyond what ablating random heads does?
- **Exploratory: Does the base decoder shape which heads matter?** Across models, do the important head positions track the decoder family, the vision encoder, or language competence?

Before answering these, we check a precondition: whether script identity (Latin vs. Devanagari) is even available in the vision encoder’s representations. If it were not, a divergence between the text and image paths would be unsurprising.

Where code-switching enters. Following [Liu et al. \(2026\)](#), we identify important heads on monolingual sentences, one language at a time, so the head sets for RQ1 come from monolingual English, Spanish, and Hindi text. Code-switched text enters in two places: the script probes, which read mixed-script images, and the causal test in RQ2, which asks whether heads identified on monolingual text change what a model produces when it reads a code-switched image. Identifying heads directly on code-switched sentences is left to future work (Section 8).

What we find. Script identity is linearly decodable from the first layer of all four vision encoders (Section 6.1). The answer to RQ1 differs by model: two models use largely separate heads for text and image input, one nearly shares them for Hindi only, and for one the measure is inconclusive. In the models that separate them, the text heads concentrate in the first two decoder layers while the image heads spread through the network (Section 6.2). For RQ2, zero-ablating the top heads for either language destroys generation in all four models, while ablating random heads does not. The heads are therefore necessary for coherent output, but the ablation cannot isolate a role in routing language (Section 6.3). For the exploratory question, the head positions shared across models are about what their layer placement alone predicts (Section 6.4).

Scope. This is an investigative study of four small VLMs, two language pairs, and one attribution method. We report what happens in each model and state how strong the evidence is for each observation (Section 7). We do not claim that the patterns generalize to other VLMs.

2 Related Work

Language-specific heads and neurons in text LLMs. Beyond LAHIS, [Tang et al. \(2024\)](#) identify language-specific neurons via parallel-corpus activation contrasts and show that deactivating them degrades only the corresponding language. [Wendler et al. \(2024\)](#) probe intermediate representations directly for an English-pivot signature. This line of work establishes that language routing in LLMs is sparse, identifiable, and at least partially causal. It is, however, entirely text-only. None of it asks what happens when the same content is presented as an image instead of a token sequence.

Code-switching circuits. Nie et al. (2025) is the closest prior work to a mechanistic study of multilingual routing failure, localizing language-confusion errors to specific token positions and late-layer transitions in text-only models. We are not aware of any prior mechanistic study of code-switching specifically (as opposed to single-language confusion), and none that touches the visual modality.

Language heads in VLMs. Ye et al. (2025) (CLAIM) is the closest prior work to ours. It identifies cross-lingual attention misalignment in VLMs and intervenes on it. The critical differences are that CLAIM studies monolingual queries in different languages rather than code-switched content, and it does not isolate the text path from the vision path, so it cannot ask whether a VLM’s *text* decoder and its *image-reading* decoder agree on which heads matter. Baek et al. (2025) characterize OCR heads in VLMs but restrict their study to English text, leaving open whether script-mixed or language-mixed visual text is routed any differently.

Reading text from pixels. Several models process text directly as rendered images, including pixel-based language models (Rust et al., 2023), a single encoder for images and rendered text (Tschannen et al., 2023), and OCR-free document and screenshot understanding (Kim et al., 2022; Lee et al., 2023). On the vision-encoder side, work on CLIP shows that image encoders represent written words separately from the scenes they appear in (Materzyńska et al., 2022), and that individual attention heads in CLIP’s image encoder specialize in specific, interpretable image properties (Gandelsman et al., 2024). Our script probes (Section 6.1) ask a narrower version of this question for the encoders inside VLMs.

Where visual information enters the decoder. Mechanistic work on VLMs has begun to trace how image information reaches and moves through the language model, using causal tracing (Palit et al., 2023; Basu et al., 2024) and by following how visual tokens are processed across layers (Neo et al., 2025). These studies localize visual information flow, but none asks whether the heads that matter for a language when it is read from an image are the same heads that matter when it is read as text.

Gap. To our knowledge, no prior work compares which attention heads a VLM relies on for the same sentence as text and as a rendered image, at head-level granularity and with the effect of the text query separated from the effect of modality, or tests those heads on code-switched images.

3 Data

All code-switched text is real and human-produced, with gold token-level language labels. We avoid synthetic code-switching, such as naive word substitution, since it risks sentences no bilingual speaker would actually write.

Hindi-English data comes from the Devanagari-script Twitter portion of the GLUECoS part-of-speech corpus (Khanuja et al., 2020), and Spanish-English data comes from the Bangor Miami corpus of transcribed bilingual speech (Deuchar et al., 2013). Monolingual baselines in English, Spanish, and Hindi come from FLORES-200 parallel triples.

Cleaning and final counts. We clean the raw social-media and transcribed-speech text with Penn Treebank detokenization, script- and noise-consistency filters, a two-stage fluency filter (heuristic checks, then Qwen2.5-0.5B perplexity, dropping the worst quartile per language pair), and a content filter for profanity and political material judged by Claude against a fixed rubric, keeping every dropped record’s drop reason for auditability.

After filtering, the corpus contains 933 code-switched sentences (806 Hindi-English from the GLUECoS training and development splits, and 127 Spanish-English), 750 monolingual sentences (250 per language), and 100 Hindi-in-Latin-script transliteration controls, which separate script identity from language identity since this content is Hindi written in the Latin alphabet.

Which data each analysis uses.

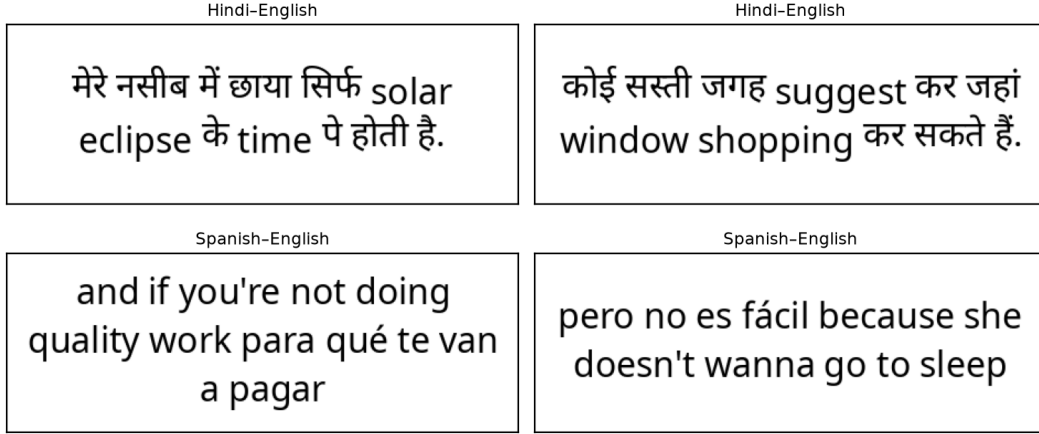


Figure 1: Four rendered code-switched sentences from the evaluation set (two Hindi-English, two Spanish-English), shown as the models see them, rendered to an image with no accompanying prompt tokens.

- **Head importance (RQ1).** For each language, 100 of the 250 monolingual FLORES sentences. Each sentence is entirely in one language, and the loss covers all of its tokens. Conditions B and C use rendered images of the same sentences. No code-switched text is used to identify heads.
- **Script probes.** The first 300 rendered code-switched images (127 Spanish-English and 173 Hindi-English), whose per-token bounding boxes give each image patch a script label. All Devanagari patches therefore come from the Hindi-English images.
- **Hindi-English ablation (RQ2).** 100 sentences sampled from the 806 Hindi-English code-switched sentences. Because heads are identified on FLORES, these images are disjoint from the data used to select heads.
- **Spanish-English ablation.** The first 100 of the 127 Spanish-English code-switched sentences, likewise disjoint from the FLORES head-scoring sentences.

Rendering. Every sentence exists as both raw text and a rendered image, black text on a white canvas, with per-token bounding boxes so patch-level attention can later be mapped back to source-language spans. Code-switched sentences render each token in a script-appropriate font, Noto Sans for Latin and Noto Sans Devanagari for Devanagari, so script boundaries are visually real rather than assumed. Figure 1 shows sample rendered stimuli. Hindi-English items mix Devanagari and Latin script within a sentence; Spanish-English items stay in Latin script throughout, which is why the script-change measure (Section 5.5) applies only to the Hindi-English pair.

4 Models

We evaluate four open VLMs, chosen to differ in vision encoder, decoder family, and image-token compression scheme while remaining small enough (1.7–3B total or decoder parameters) to run full LAHIS gradient attribution (forward and backward passes with a per-head soft mask) on the hardware available to us, a single NVIDIA T4 GPU with 16GB of VRAM. Model size was chosen to fit this budget, not for any property of the models themselves.

- **Qwen2-VL-2B-Instruct** (Wang et al., 2024) pairs a custom 2D-RoPE vision transformer (0.67B) with a 2×2 patch-merger and a Qwen2-1.5B decoder (28 layers, 12 attention heads).
- **InternVL3-2B** (Chen et al., 2024) pairs InternViT-300M (v2.5), with pixel-shuffle compression, and a Qwen2.5-1.5B decoder (28 layers, 12 attention heads).

Table 1: Decoder language competence on monolingual FLORES text. Per-token perplexity is deflated for Hindi by tokenizer fertility and is not comparable across languages. The character-normalized values are comparable and show Hindi is hardest for every model, with all models in range. PaliGemma-3B is scored in its native prefix-LM regime.

Model	Per-token ppl.			Char.-normalized ppl.		
	en	es	hi	en	es	hi
Qwen2-VL-2B	40.9	26.2	5.9	2.06	2.23	4.46
InternVL3-2B	38.4	28.0	6.7	2.04	2.28	4.87
SmolVLM	83.8	49.5	7.4	2.23	2.98	7.64
PaliGemma-3B	91.3	114.7	69.4	2.48	2.75	4.48

- **SmolVLM-Instruct** (Marafioti et al., 2024) pairs a SigLIP-SO400M (Zhai et al., 2023) vision encoder and a fixed-scale-factor pixel-shuffle connector with a SmolLM2-1.7B decoder (24 layers, 32 attention heads).
- **PaliGemma-3B (mix-448)** (Beyer et al., 2024) pairs a SigLIP-SO400M vision encoder at 448px and a linear projector that keeps all 1024 image tokens (no merging or pooling) with a Gemma-2B decoder (18 layers, 8 attention heads, multi-query attention). It is a *prefix-LM*, so image and prompt tokens receive bidirectional attention and only the suffix is trained causally. We therefore score its decoder in that native suffix regime, taking the loss on the text tokens of `[BOS] \n text`, for both the competence gate and the LAHIS A condition.

Qwen2-VL-2B and InternVL3-2B share a decoder family (Qwen2 and Qwen2.5). To our knowledge, Qwen2.5 is a separate pretraining run in that family rather than a continuation of Qwen2’s weights (Qwen Team, 2024), so the two decoders share architecture and training recipe but not weights. SmolVLM’s SmolLM2 decoder is Llama-architecture and PaliGemma’s is Gemma, each from an unrelated pretraining lineage. The exploratory analysis in Section 6.4 uses this structure.

Language competence gate. Before running any language-routing analysis, we verify that each extracted decoder is competent in each language, so that head-importance scores do not measure noise. We compute decoder perplexity on monolingual FLORES text (Table 1), using a per-token threshold of 100. All four models pass, though PaliGemma-3B’s per-token Spanish (114.7) slightly exceeds it.

Per-token perplexity must not be compared across languages, because tokenizer fertility differs by 3.4 to 3.6 times between English (about 0.23 tokens per character) and Devanagari Hindi (about 0.78). This mechanically deflates Hindi’s per-token perplexity regardless of competence. We therefore normalize by characters, using $\text{ppl}_c = \exp(\text{NLL}/n_{\text{chars}})$. On this scale Hindi is the hardest language for every model, and all cross-language competence statements in this paper use the character-normalized values.

5 Method

Figure 2 gives an overview of this pipeline before we describe each part in detail.

5.1 Three conditions

For each language and each sentence, we evaluate three conditions:

- A (text)** The extracted decoder processes the raw sentence text. This decoder is the language model sub-module taken *from inside* the VLM after its own vision-language instruction tuning, not a separately loaded checkpoint. The vision encoder never runs.
- B (image+query)** The full VLM processes the rendered image plus a fixed textual instruction (“Read this:”), with the loss computed on the appended target sentence.

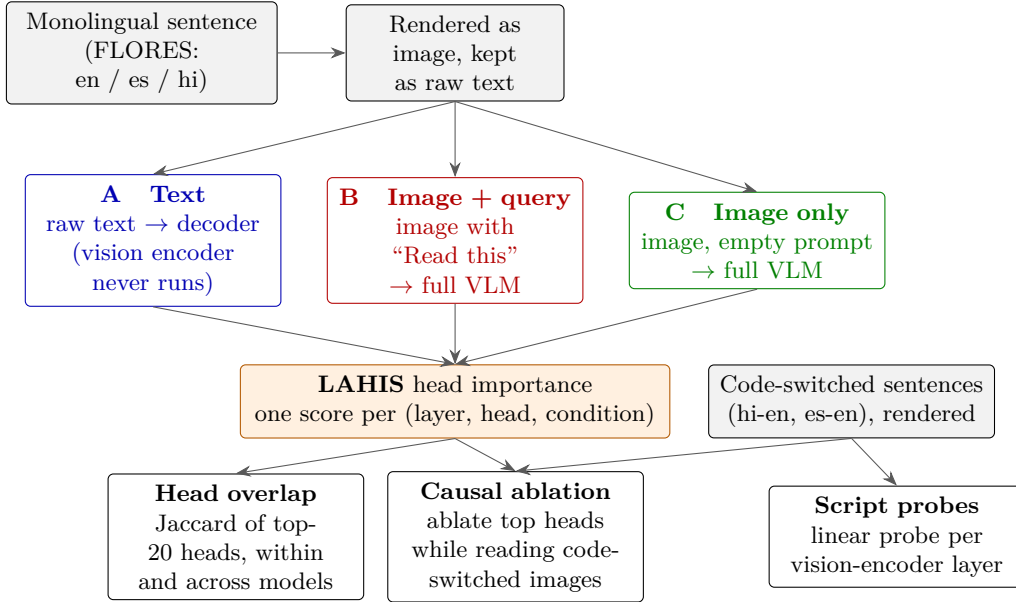


Figure 2: Overview of the pipeline. Head importance is scored with LAHIS on monolingual sentences in three conditions (A, B, C), and the top heads are compared across conditions and models. Code-switched sentences, rendered as images, are used for the script probes and for the causal ablation, which removes the top heads while the model reads them.

C (image-only) Identical to B but with an empty instruction. It uses the same processor chat template as B (image block, user and assistant role tokens), with the user’s text field left empty, so it adds no instruction-language tokens beyond the role tokens that B and C already share.

Comparing B with C isolates the effect of the text query, and comparing A with C isolates the effect of modality. Using the decoder *extracted from* the VLM for condition A, rather than the original pre-finetuning checkpoint, is essential. It ensures that A vs. C isolates modality alone, without confounding modality with whether the weights have seen vision-instruction tuning at all.

5.2 Head importance (LAHIS)

Following Liu et al. (2026), we install a differentiable soft mask $M \in \mathbb{R}^{L \times H}$ (initialized to all ones) that multiplies each attention layer’s per-head output via a forward pre-hook on the output projection. This works because the projection’s *input* is the concatenation of per-head outputs and can therefore be sliced exactly per head (unlike hooking the post-projection output, which mixes heads). We compute the language-modeling loss and take $|\partial \text{loss} / \partial M|$ as the importance score for each head, averaged over 100 monolingual FLORES sentences per condition per language, with the loss taken over all tokens of each sentence (Section 3). This requires only one forward and one backward pass per sentence. We visualize these scores as layer×head maps in Appendix E, for Hindi (Figure 7) and Spanish (Figure 8).

What this score measures. The loss behind LAHIS is the ordinary next-token loss, not a loss tied to language identity. A head scores highly if removing it hurts prediction of text in the given language, which includes heads that do general work such as tracking position or copying earlier tokens. We therefore read a high score as “important for processing this language”, not “specific to this language”. Section 6.4 shows that most top-scoring heads are indeed shared across all three languages within each model.

5.3 Head overlap

For each model and language, we compute the top-20 heads by importance under each condition and report pairwise Jaccard overlap ($|A \cap B|/|A \cup B|$) between conditions. We additionally compute a 10,000-sample permutation baseline (randomly-drawn 20-head sets from the same layer $\times$ head grid, computed separately per model since the grids differ in size) to establish chance-level overlap, and sweep $K \in \{5, 10, 20, 40, 80\}$ to check that our results are not an artifact of the specific top-20 cutoff.

For the exploratory cross-model analysis, we additionally record the raw (layer, head) coordinates in each model’s top-20 text-condition set and compute pairwise coordinate overlap between every pair of models.

5.4 Script probes

To ask whether script identity is already linearly decodable inside the vision encoder, before any information reaches the decoder, we train balanced-accuracy logistic-regression probes (5-fold cross-validated) at every vision-encoder layer, predicting each patch’s script (Latin, Devanagari, or background) from its frozen hidden state. Patch-to-script labels come from the token bounding boxes recorded at render time, mapped onto the actual grid of image tokens the decoder sees.

This grid differs from the raw patch grid whenever a compression module, such as a patch merger or pixel-shuffle layer, runs between encoder and decoder. Background patches are labeled only to be excluded. Every patch we actually fit or score the probe on is Latin or Devanagari, so the reported classifier is binary and chance is 0.5, not 1/3. We keep a random 100 patches per image and 4,000 text patches per layer, and fit a class-balanced logistic regression evaluated by stratified 5-fold cross-validation. The folds split patches rather than images, so patches from one image can appear in both training and test folds (Section 8).

5.5 Causal ablation

Head importance from gradient attribution is correlational. A head can receive a large gradient without being causally responsible for the model’s output. We test causality directly by zero-ablating the top-20 heads found under condition A for English and for Hindi separately, via the same per-head output-projection hook, now clamping the targeted heads’ output to zero, and generating text from each of 100 Hindi-English code-switched images (Section 3). Generation is greedy and deterministic (no sampling, 64-token budget), so ablation variants differ only in which heads are masked, not in decoding randomness.

We record whether the output script changes from the un-ablated baseline. We restrict this test to Hindi-English images because the outcome measure is a script change, and script is only informative when the two code-switched languages do not share one. Spanish-English is handled separately with a language-identification metric (Table 9).

The ablated sets are simply the 20 highest-scoring heads for each language, not heads that score higher for one language than the other. Because most top heads are shared across languages (Section 6.4), the English and Hindi sets overlap heavily, so “Hindi heads” and “English heads” name the language the heads were scored on rather than heads specific to that language.

We compare against two controls. The first is a random 20-head set, drawn from heads *outside* the top-20 English and Hindi sets and redrawn for each seed, to test whether *any* 20-head ablation changes output this much. The second is a mean-ablation variant, which replaces each head’s output with its mean taken over token positions within that same image’s baseline forward pass (per-sentence, per-head, not corpus-wide) rather than zero, to separate language-related suppression from generic degradation.

Script labels and fluency. Each output gets a script label (Latin, Devanagari, other, or none when it contains no alphabetic characters), and any change from the un-ablated label counts as a script change. Because an output with no letters also changes the label, we separately check fluency: an ablated output counts as fluent if it keeps at least half of the words of the model’s un-ablated output for the same image. This is a rough proxy, so we read it against the random-head control rather than as an absolute measure.

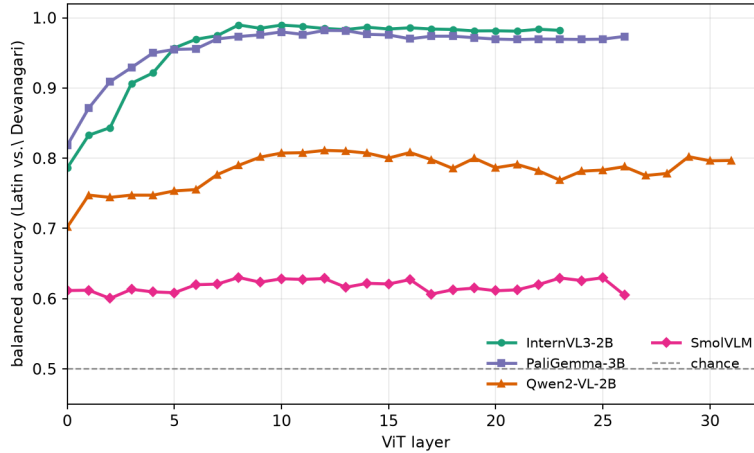


Figure 3: Script-probe balanced accuracy (Latin vs. Devanagari patches, 5-fold CV) at every vision-encoder layer, all four models (dashed line: chance, 0.5). InternVL3-2B and PaliGemma-3B both rise quickly and saturate near 1.0 (0.990 and 0.982 peak). Qwen2-VL-2B rises more slowly to a lower plateau (0.811). SmolVLM stays close to its layer-0 value throughout (0.630 peak), so its weak script separability is not a matter of needing more depth. PaliGemma and SmolVLM share the SigLIP-SO400M encoder family yet sit at opposite extremes (Section 6.1).

6 Results

6.1 Precondition: script identity is decodable early in every vision encoder

Script-probe balanced accuracy exceeds the 50% chance level even at layer 0 for all four encoders, confirming script identity is present in the earliest patch embeddings before any contextualization across the image.

The four encoders differ substantially in how well. InternVL3-2B’s InternViT reaches 0.990 balanced accuracy by layer 8 of 24 and stays above 0.98 through its final layer, near-perfect linear separability. PaliGemma-3B’s SigLIP-SO400M encoder is nearly as strong, 0.982 at layer 12 of 27, climbing from an already-high 0.818 at layer 0. Qwen2-VL’s custom encoder peaks lower, 0.811 at layer 12 of 32. SmolVLM’s SigLIP-SO400M encoder is the weakest by a wide margin and, unlike the other three, never improves substantially with depth. It peaks at only 0.630 (layer 8 of 27) and *declines* to 0.605 by its final layer.

PaliGemma-3B and SmolVLM share the same vision-encoder family (SigLIP-SO400M), yet sit at opposite extremes, 0.982 against 0.630. The two deploy it differently. PaliGemma runs SigLIP at 448px and keeps all 1024 patch tokens, whereas SmolVLM feeds it lower-resolution, pooled tokens. Within one encoder family, then, the script signal varies with resolution and token budget. All four models are given the same rendered images with the same fonts, so this spread comes from the encoders rather than the stimuli. Figure 3 shows the full per-layer trajectories.

These probes track rendered script rather than language content. We apply each model’s already-trained, peak-layer Latin-versus-Devanagari probe to the 100 Hindi-in-Latin-script transliteration controls (Section 3), which are Hindi *words* written in Latin *glyphs*. The probe calls them Latin at a rate matching its own code-switched-data accuracy (InternVL3-2B 99.7%, PaliGemma-3B 99.7%, Qwen2-VL-2B 97.3%, SmolVLM 84.0%, against 98.8%, 98.2%, 80.2%, and 62.4% balanced accuracy on real script respectively). The probes read orthography, not the underlying language. Because this check applies trained probes to images they never saw, it is not affected by the patch-level cross-validation noted in Section 5.4.

Script identity is therefore available to every decoder in the study. Where a model’s text and image paths use different heads (Section 6.2), it is not because the encoder failed to separate the scripts.

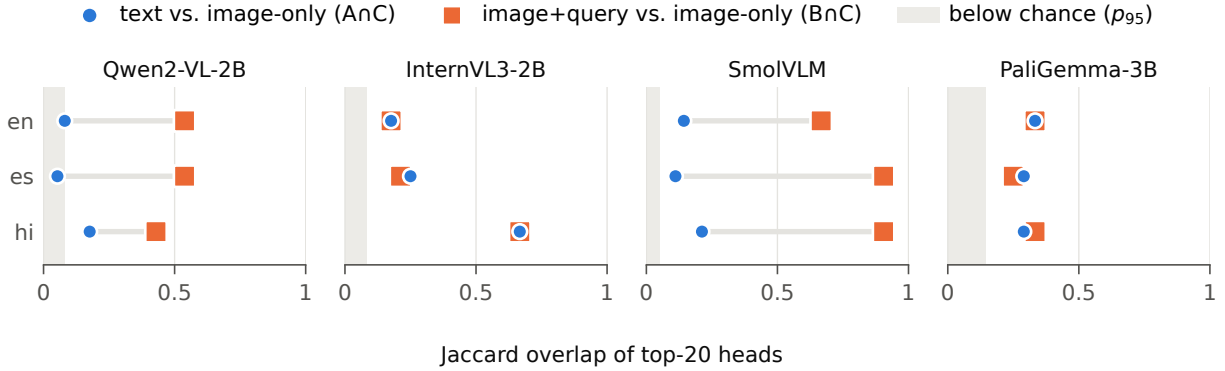


Figure 4: Do text and image inputs recruit the same heads? Each row compares text-vs-image overlap ($A \cap C$, circles) with image-vs-image overlap ($B \cap C$, squares) for one language. A long bar means the image conditions agree with each other but not with text. Where the markers coincide, text and image agree as much as the two image conditions do. The shaded band is each model’s chance level (p_{95} : 0.081 for Qwen2-VL-2B and InternVL3-2B, 0.053 for SmolVLM, 0.143 for PaliGemma-3B). Values are in Table 4.

6.2 RQ1: whether text and image inputs share heads depends on the model

Figure 4 plots, for each model and language, how much the top-20 heads for text overlap with those for the image alone ($A \cap C$), next to how much the two image conditions overlap with each other ($B \cap C$), against a permutation-derived chance level (95th percentile over 10,000 random 20-head draws). Table 4 (Appendix A) gives all three pairwise overlaps. The four models show three different patterns.

Separate heads: Qwen2-VL-2B and SmolVLM. In both models, text-image overlap ($A \cap C$) stays low across every language tested (0.05–0.21), while the two image conditions cluster tightly together ($B \cap C$ 0.43–0.91, reaching its highest values for SmolVLM). Once an image is present, these models recruit a consistent head set whether or not a text instruction accompanies it, and that set differs substantially from the heads that matter for the same sentence as text. The LAHIS maps make the difference visible: Qwen2-VL-2B’s text-condition importance sits in layers 0–1, while its image conditions draw on heads throughout the network (Figure 7).

Nearly shared heads, for Hindi only: InternVL3-2B. InternVL3-2B’s Hindi $A \cap C$ reaches 0.667 and $A \cap B$ reaches 0.905, close to the same head set for Hindi as tokens and as pixels. Its English and Spanish overlaps are lower and pattern more like the other models, though still generally higher. This does not follow from better Hindi competence: InternVL3-2B’s character-normalized Hindi perplexity (4.87) is no better than Qwen2-VL-2B’s (4.46), whose heads do not overlap. With one model and one language, we treat this as an observation to be explained rather than a pattern.

Inconclusive: PaliGemma-3B. PaliGemma-3B’s overlaps fall in a narrow 0.21–0.33 band only modestly above its elevated chance level (0.143, a consequence of its small 18×8 head grid), and its image-image overlap ($B \cap C$ 0.25–0.33) is not distinctly higher than its text-image overlaps. The measure cannot tell whether this model shares heads across modalities or not. With 144 heads in total, a top-20 set is too large a fraction of the grid for Jaccard overlap to resolve the difference.

Where the top heads sit. Within one model, heads share a numbering, so the overlap can also be read by layer (Figure 5; counts in Table 5). In Qwen2-VL-2B, 19–20 of the top 20 text-condition heads lie in layers 0–1, against 2–6 of the image-condition heads, which is why its text and image sets barely intersect. InternVL3-2B also places its text-condition heads early (14–18 of 20), and for Hindi its image-condition heads follow them there (17 and 14), matching its near-shared Hindi heads; for English its image-only heads do

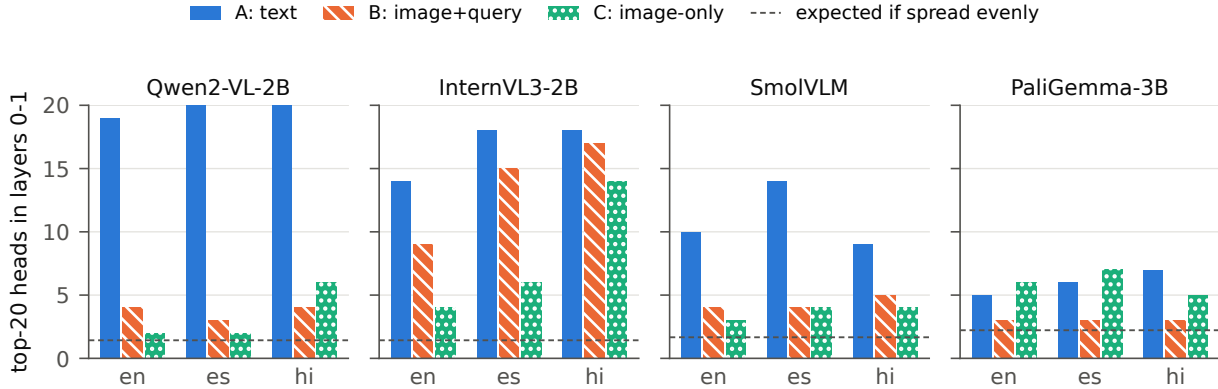


Figure 5: Where do the top heads sit? Number of each top-20 set that lies in decoder layers 0–1, by model, language, and condition. The dashed line is the count expected if the 20 heads were spread evenly over the model’s heads (1.4 for the two Qwen-family models, 1.7 for SmolVLM, 2.2 for PaliGemma-3B). Counts are in Table 5.

not (4). SmolVLM’s text-condition heads are partly early (9–14) and its image-condition heads mostly not (3–5). PaliGemma-3B places only 3–7 heads there in every condition, close to what an even spread would give.

Robustness to K . Sweeping $K \in \{5, 10, 20, 40, 80\}$ (Appendix C), text-image overlap ($A \cap B$, $A \cap C$) stays below image-image overlap ($B \cap C$) for Qwen2-VL-2B and SmolVLM at every K , and InternVL3-2B’s three Hindi conditions stay close together (0.58–1.0) at every K . The three patterns are not an artifact of the $K=20$ cutoff.

Encoder signal and shared heads. The two models with the strongest encoder script signal behave differently. InternVL3-2B (0.990) nearly shares heads for Hindi, while for PaliGemma-3B (0.982) the overlap measure is inconclusive. A strong script signal in the encoder does not, on its own, produce shared text and image heads in these models, though PaliGemma’s inconclusive overlap means this comparison rests mainly on the contrast between InternVL3-2B and the others.

6.3 RQ2: removing the top heads breaks generation, so ablation cannot isolate language routing

We first scored each ablation by whether the output’s script changed from the un-ablated baseline. Inspecting the outputs shows that this mostly records something else: zero-ablating the top-20 heads for *either* language destroys generation in every model. Figure 6 shows the share of outputs that remain fluent (Section 5.5), pooled over a single run and three seeds (400 generations per cell, 100 images each).

The top heads are necessary for coherent output. With the top-20 Hindi or English heads zero-ablated, at most 4 of 400 outputs remain fluent in any model. With 20 random heads ablated instead, 283 to 303 of 400 do. The failures take different forms. PaliGemma-3B returns an empty string for 389 of 400 Hindi-head ablations. Qwen2-VL-2B returns letterless output (runs of newlines, ellipses, or backticks) for 278 of 400 Hindi-head ablations and CJK fragments for 328 of 400 English-head ablations. InternVL3-2B’s Hindi-head ablations contain CJK characters in all 400 generations, and SmolVLM emits repeated `<row_2>` tokens. No un-ablated output in any model contains CJK characters. Mean ablation breaks generation just as thoroughly, with one exception: SmolVLM keeps 243 of 400 outputs fluent when its Hindi heads are mean-ablated.

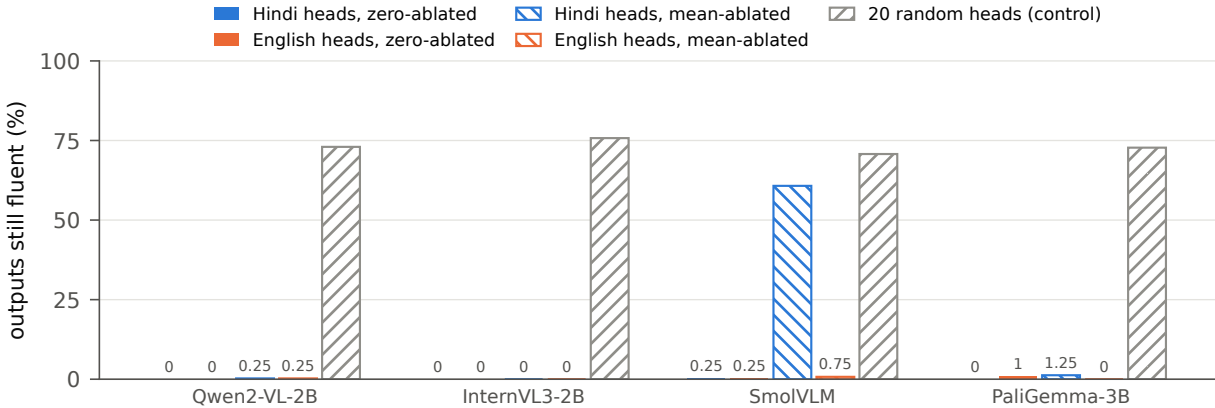


Figure 6: Does output survive ablation? Percent of Hindi-English code-switched generations that remain fluent (keep at least half of the un-ablated output’s words), pooled over four runs of 100 images. Zero-ablating the top-20 heads for either language leaves at most 1% fluent in every model, against 71–76% for 20 random heads. Mean ablation breaks output as well, except for SmolVLM’s Hindi heads. Small values are printed above their bars.

What this does and does not show. The heads LAHIS ranks highest on text are necessary for the models to produce coherent output when reading code-switched images, and a random set of the same size is not. The ablations do not show that these heads route language or script. Removing the Hindi heads and removing the English heads cause the same collapse, and the two sets largely coincide (Section 6.4). In Qwen2-VL-2B they occupy 19–20 of the 24 heads in layers 0–1 (Figure 5), so the collapse may be generic damage to the first layers rather than anything specific to language. Testing language routing would need a random control drawn from the same layers, and head sets chosen by contrasting one language’s scores with the other’s.

Why script-change rates mislead here. Letterless output changes the script label, so collapse registers as a script change. In the single run, Qwen2-VL-2B’s 74% Hindi-head script-change rate consists of 69 Latin and 5 Devanagari outputs that became letterless, and PaliGemma-3B’s 97% of 82 and 15. InternVL3-2B’s rate is low (14%) because its collapsed outputs mix CJK characters with enough Latin letters to keep a Latin label; in every run its changed items include all of its baseline-Devanagari outputs, and SmolVLM’s changed items are exactly its baseline-Devanagari outputs. The apparent differences between models in script change, including InternVL3-2B’s lack of an effect and the differences between mean and zero ablation, therefore reflect how each model fails rather than how it routes language. The script-change tables are in Appendix D for completeness. The Spanish-English ablation, scored by running a language identifier on the output (Table 9), has not been checked for collapse and is likely affected in the same way.

Baseline script differences. Without any ablation, SmolVLM produces Devanagari script far more often than the other models (37 of 100 outputs, against 5 for Qwen2-VL-2B, 11 for InternVL3-2B, and 15 for PaliGemma-3B). This is not explained by competence: SmolVLM is the weakest model on Hindi by character-normalized perplexity (Table 1). We report it as a difference in decoding behavior that we do not explain.

6.4 Exploratory: shared head positions across models reflect layer placement

Within each model, most top text-condition heads are shared across languages: 18 heads are in the top 20 for all three languages in Qwen2-VL-2B, 14 in InternVL3-2B, 14 in SmolVLM, and 15 in PaliGemma-3B. Most heads that LAHIS ranks highly are therefore language-general, consistent with what the score measures (Section 5.2), and the “Hindi” and “English” sets ablated in Section 6.3 are largely the same heads.

Table 2: Shared (layer, head) positions between two models’ top-20 text-condition heads, per language. The Qwen-pair expectation assumes each model’s heads in layers 0–1 are placed at random within those two layers.

Pair	en	es	hi
Qwen2-VL-2B – InternVL3-2B	13	16	15
expected from layer placement	≈ 11	15	15
Qwen2-VL-2B – PaliGemma-3B	5	6	7
InternVL3-2B – PaliGemma-3B	5	6	7
Qwen2-VL-2B – SmolVLM	2	2	1
InternVL3-2B – SmolVLM	1	1	1
SmolVLM – PaliGemma-3B	0	0	0

Across models, Table 2 counts the (layer, head) positions that two models’ top-20 text-condition sets share, per language. The two Qwen-family models share 13, 16, and 15 positions for English, Spanish, and Hindi. PaliGemma-3B shares 5–7 with each of them, SmolVLM shares 1–2 with them, and SmolVLM and PaliGemma-3B share none.

Layer placement accounts for most of this pattern. Both Qwen-family models put most of their top text-condition heads in layers 0–1 (Qwen2-VL-2B 19, 20, and 20 of 20; InternVL3-2B 14, 18, and 18), which hold 24 heads in each model. If each model’s heads were placed at random within those two layers, the two sets would share about 11, 15, and 15 positions, against the observed 13, 16, and 15. The other models place fewer top heads in layers 0–1 (SmolVLM 9–14, PaliGemma-3B 5–7) and have differently shaped grids, so they match fewer positions.

Beyond layer placement, a matching position carries little meaning. Heads within a layer are interchangeable across independently trained models, and none of our decoders share weights, including Qwen2 and Qwen2.5 (Section 4). What these comparisons show is a layer-level pattern: the top text-condition heads of the Qwen-family models sit in the first two layers, SmolVLM’s partly, and PaliGemma-3B’s mostly elsewhere. Whether that pattern follows decoder family cannot be tested with four models whose decoder family also varies with head count, grid shape, attention scheme, and image compression.

7 What we can and cannot conclude

Table 3 summarizes each observation, the evidence behind it, and how strongly we think that evidence supports it. *Supported* means consistent evidence with a matched control; *tentative* means one model, one language, or overlapping intervals; *descriptive* means a pattern we report but cannot test with this design.

Two practical points follow even from the more cautious rows. First, one VLM’s language heads cannot be assumed to be another’s: the four models differ in whether text and image inputs share heads, in where their top heads sit, and in how strongly their encoders separate scripts. Second, ablation results need an output-quality check. In our study, a script-change metric registered collapsed output as a change of script and produced differences between models that disappear once fluency is checked.

8 Limitations

Model scale and number. All four models are in the 1.7–3B range, a compute constraint rather than a scientific choice. Whether any of the patterns hold at 7–8B scale, or in other small VLMs, is untested.

Heads are identified on monolingual text. The head sets come from monolingual sentences, so RQ1 compares text and image processing of single-language content, and code-switching enters only through the probes and the ablation test. Scoring heads on the code-switched sentences themselves, for example with the loss restricted to Hindi or English tokens using the gold token labels, would tie the head sets to code-switched input directly.

Table 3: Summary of observations and the strength of evidence for each, within the four models studied.

Observation	Evidence	Strength
Script identity is linearly decodable from the first vision-encoder layer, in all four models	Per-layer probes; out-of-sample transliteration check; patch-level folds may inflate accuracy (Section 6.1)	Supported
Qwen2-VL-2B and SmolVLM use largely different heads for text and image input	Jaccard overlap above chance; stable across K (Section 6.2)	Supported (single run)
InternVL3-2B uses nearly the same heads for Hindi text and Hindi images	One model, one language; its Hindi image heads sit in the same early layers (Section 6.2)	Tentative
PaliGemma-3B shares or separates heads	Overlap near its chance level (Section 6.2)	Inconclusive
The top-20 heads are necessary for coherent output when reading code-switched images	Zero ablation leaves $\leq 1\%$ fluent vs. 71–76% for random heads, all four models; rough fluency proxy (Section 6.3)	Supported
These heads route output language or script	Heads of both languages cause the same collapse; no layer-matched control (Section 6.3)	Not supported
Shared head positions across models follow decoder family	Same-family matches are about what layer placement predicts (Section 6.4)	Not supported beyond layer placement

What LAHIS and the ablation measure. LAHIS ranks heads by their effect on next-token loss, so its top heads include language-general ones (Section 5.2), and the English and Hindi top-20 sets largely coincide. The ablations use these absolute top-20 sets rather than heads contrasted between languages, draw the random control from the whole network rather than from the same layers, and judge fluency with a rough word-overlap proxy. Together these mean the ablations can show that the heads are necessary for coherent output, but not what role they play in routing language.

Probe evaluation. The script probes’ cross-validation folds split patches rather than images, so patches from one image can appear in both training and test folds, which likely inflates the reported accuracies. The transliteration check applies trained probes to separate images and is not affected. Re-running with folds grouped by image would give a cleaner estimate. All Devanagari patches also come from the 173 Hindi-English images among the 300 probed.

Run-to-run stability. The ablations are repeated across three seeds, each with a fresh sample of 100 sentences and a fresh random-head draw. The head-overlap values are single-run, controlled only by the permutation baseline.

Covarying architecture. Decoder family covaries with head-grid shape, head count, attention scheme, and image compression, none of which we vary independently. PaliGemma-3B’s small 18×8 grid also raises its chance overlap (Table 4).

Two language pairs. With only Hindi-English and Spanish-English, we cannot separate script distance from language-family distance as the driver of InternVL3-2B’s Hindi-specific overlap. The corpus is also imbalanced (806 vs. 127), though LAHIS uses a matched per-language sample, so this affects only the diversity of the Spanish-English ablation stimuli.

9 Conclusion

We investigated whether small VLMs rely on the same attention heads for a language when it arrives as text tokens and as a rendered image, and whether those heads matter when the models read code-switched text.

In the four models we studied, the answer depends on the model. Two use largely different heads for the two inputs, with the text heads concentrated in the first decoder layers; one uses nearly the same heads but only for Hindi; and for one our measure cannot tell. Every vision encoder makes script identity available from its first layer, so these differences do not come from the encoders failing to separate scripts. Removing the top heads destroys generation in all four models while random heads do not, which shows the heads are necessary but not that they route language. Our first measure of this ablation, a script-change rate, counted the collapse as a change of script, a failure mode other head-ablation studies should check for. Future work could identify heads on code-switched text directly, contrast heads between languages, use layer-matched ablation controls with a proper fluency measure, and extend the analysis to larger VLMs and to language pairs that separate script from language family.

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A Full Overlap Values

Table 4: Jaccard overlap of top-20 heads across conditions. A = text, B = image+query, C = image-only. Lang is the language of the monolingual FLORES sentences used to compute the LAHIS scores (Section 3). Chance (p_{95}): 0.081 (Qwen2-VL-2B, InternVL3-2B), 0.053 (SmolVLM), 0.143 (PaliGemma-3B, whose small 18×8 head grid raises its chance level).

Model	Lang	$A \cap B$	$A \cap C$	$B \cap C$
Qwen2-VL-2B	en	0.143	0.081	0.538
	es	0.081	0.053	0.538
	hi	0.111	0.176	0.429
InternVL3-2B	en	0.333	0.176	0.176
	es	0.538	0.250	0.212
	hi	0.905	0.667	0.667
SmolVLM	en	0.176	0.143	0.667
	es	0.111	0.111	0.905
	hi	0.250	0.212	0.905
PaliGemma-3B	en	0.250	0.333	0.333
	es	0.212	0.290	0.250
	hi	0.250	0.290	0.333

B Top Heads in Layers 0–1

Table 5: Number of each top-20 set that lies in decoder layers 0–1, by model, language, and condition (the values plotted in Figure 5). Layers 0–1 hold 24 heads in Qwen2-VL-2B and InternVL3-2B, 64 in SmolVLM, and 16 in PaliGemma-3B.

Model	Lang	A: text	B: image+query	C: image-only
Qwen2-VL-2B	en	19	4	2
	es	20	3	2
	hi	20	4	6
InternVL3-2B	en	14	9	4
	es	18	15	6
	hi	18	17	14
SmolVLM	en	10	4	3
	es	14	4	4
	hi	9	5	4
PaliGemma-3B	en	5	3	6
	es	6	3	7
	hi	7	3	5

C K -Sweep

Table 6 shows $A \cap C$ and $B \cap C$ at the two extremes of our $K \in \{5, 10, 20, 40, 80\}$ sweep, bracketing the $K=20$ values in Table 4. The full sweep (all five K values, all four models, all three languages) is released with the code as `checkpoints/head_overlap_matrix.json`.

Table 6: Text-image ($A \cap C$) vs. image-image ($B \cap C$) Jaccard overlap at $K=5$ and $K=80$, the two extremes of the sweep. Qwen2-VL-2B and SmolVLM keep $A \cap C$ well below $B \cap C$ at both extremes, for every language. InternVL3-2B’s Hindi row keeps $A \cap C$ and $B \cap C$ close together at both extremes. PaliGemma-3B’s overlaps sit uniformly higher ($A \cap C$ already 0.25 at $K=5$) because its small 18×8 head grid inflates chance-level Jaccard (Table 4); its $A \cap C$ and $B \cap C$ track each other loosely at both extremes, consistent with the inconclusive result at $K=20$.

Model	Lang	$A \cap C@5$	$A \cap C@80$	$B \cap C@5$	$B \cap C@80$
Qwen2-VL-2B	en	0.000	0.240	0.429	0.539
	es	0.000	0.176	0.429	0.553
	hi	0.111	0.240	0.667	0.650
InternVL3-2B	en	0.000	0.301	0.111	0.468
	es	0.000	0.301	0.000	0.416
	hi	1.000	0.584	1.000	0.739
SmolVLM	en	0.000	0.231	1.000	0.861
	es	0.111	0.203	1.000	0.839
	hi	0.000	0.368	1.000	0.882
PaliGemma-3B	en	0.250	0.495	0.429	0.778
	es	0.250	0.538	0.111	0.616
	hi	0.250	0.468	0.250	0.667

D Script-Change Rates

These tables report the script-change rates we first used to score the ablations. As Section 6.3 explains, nearly all zero-ablation changes are collapsed output rather than output in the other script, so these rates do not measure language routing. Table 7 is a single run (random seed 42) and Table 8 averages three seeds, each drawing its own 100 of the 806 Hindi-English sentences and its own random-head control. Table 9 gives the Spanish-English rates, scored with a language identifier (`langid`, es vs. en) on the generated text.

Table 7: Single run (random seed 42): % of 100 Hindi-English code-switched generations whose output script label changes from the un-ablated baseline, by ablation variant (top-20 heads for the named language). The four columns are Qwen2-VL-2B, InternVL3-2B, SmolVLM, and PaliGemma-3B. Nearly all zero-ablation changes are collapsed output (Section 6.3). PaliGemma-3B’s random-head draw in this run changed 31% of outputs, against 5–11% in the three reseeded draws; about half of its generations were degenerate, suggesting it hit heads the model depends on.

Variant	Qwen	Intern	Smol	PaliG
Zero-ablate en	53%	25%	37%	18%
Zero-ablate hi	74%	14%	37%	97%
Mean-ablate en	33%	20%	52%	61%
Mean-ablate hi	29%	24%	18%	30%
Random-20 (control)	9%	9%	9%	31%

Table 8: Hindi-English script-change rates (percent) as mean $\pm$ standard deviation across three seeds. hi-/en-zero and hi-/en-mean zero- and mean-ablate the top-20 heads scored on Hindi or English; random ablates 20 random heads outside those sets, redrawn per seed.

Model	hi-zero	en-zero	random	hi-mean	en-mean
Qwen2-VL-2B	70.3 $\pm$ 5.0	46.3 $\pm$ 3.4	7.0 $\pm$ 4.9	30.3 $\pm$ 5.4	33.7 $\pm$ 8.8
InternVL3-2B	11.7 $\pm$ 1.2	17.0 $\pm$ 2.2	14.0 $\pm$ 6.2	22.7 $\pm$ 3.3	23.3 $\pm$ 2.5
SmolVLM	35.3 $\pm$ 2.1	35.3 $\pm$ 2.1	8.7 $\pm$ 0.9	23.3 $\pm$ 3.3	47.0 $\pm$ 1.6
PaliGemma-3B	97.3 $\pm$ 0.5	15.3 $\pm$ 2.9	7.0 $\pm$ 2.8	29.3 $\pm$ 2.6	65.0 $\pm$ 2.9

Table 9: Spanish-English ablation. Percent of 100 es-en generations whose dominant language (`langid`, es vs. en) changes from the un-ablated baseline, zero-ablating the top-20 heads for Spanish or for English, against the random-head control. The control is raised by language-ID noise on short generations. These rates have not been checked for collapsed output.

Model	es-heads	en-heads	Random-20
Qwen2-VL-2B	17%	14%	8%
InternVL3-2B	48%	53%	33%
SmolVLM	39%	51%	8%
PaliGemma-3B	85%	82%	19%

E LAHIS Head-Importance Maps

Figure 7 shows the raw per-head LAHIS importance behind the overlap analysis, for Hindi, as a layer $\times$ head grid per model and condition. Qwen2-VL-2B’s text importance (A) is confined to layers 0–1, while its image conditions (B, C) recruit heads throughout the depth, so the top-20 sets barely intersect. InternVL3-2B’s A, B and C light up nearly the same early heads. SmolVLM’s importance is sparse and scattered across its wide 24×32 grid. PaliGemma-3B’s importance is spread diffusely across layers with no sharp concentration, so its top-20 sets overlap only at the chance level its small 18×8 grid dictates.

LAHIS per-head importance (Hindi), top-20 heads ringed

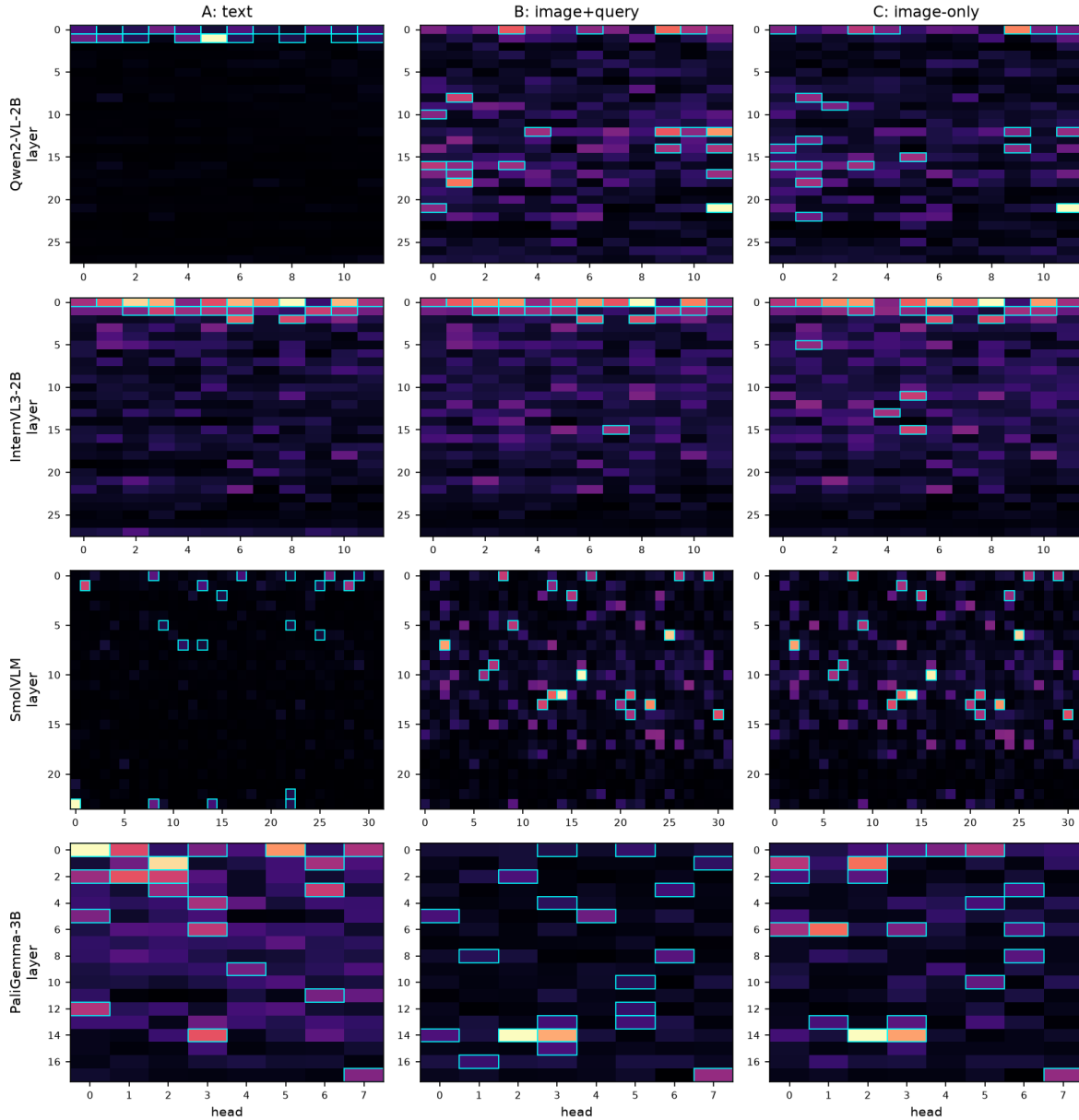


Figure 7: LAHIS per-head importance for Hindi, min-max normalized within each panel. Rows are the four models and columns the three routing conditions (A=text, B=image+query, C=image-only). Cyan rings mark each panel’s top-20 heads (the set entering the overlap analysis). Grid shapes differ by model (28×12, 28×12, 24×32, 18×8). Brighter is more important.

LAHIS per-head importance (Spanish), top-20 heads ringed

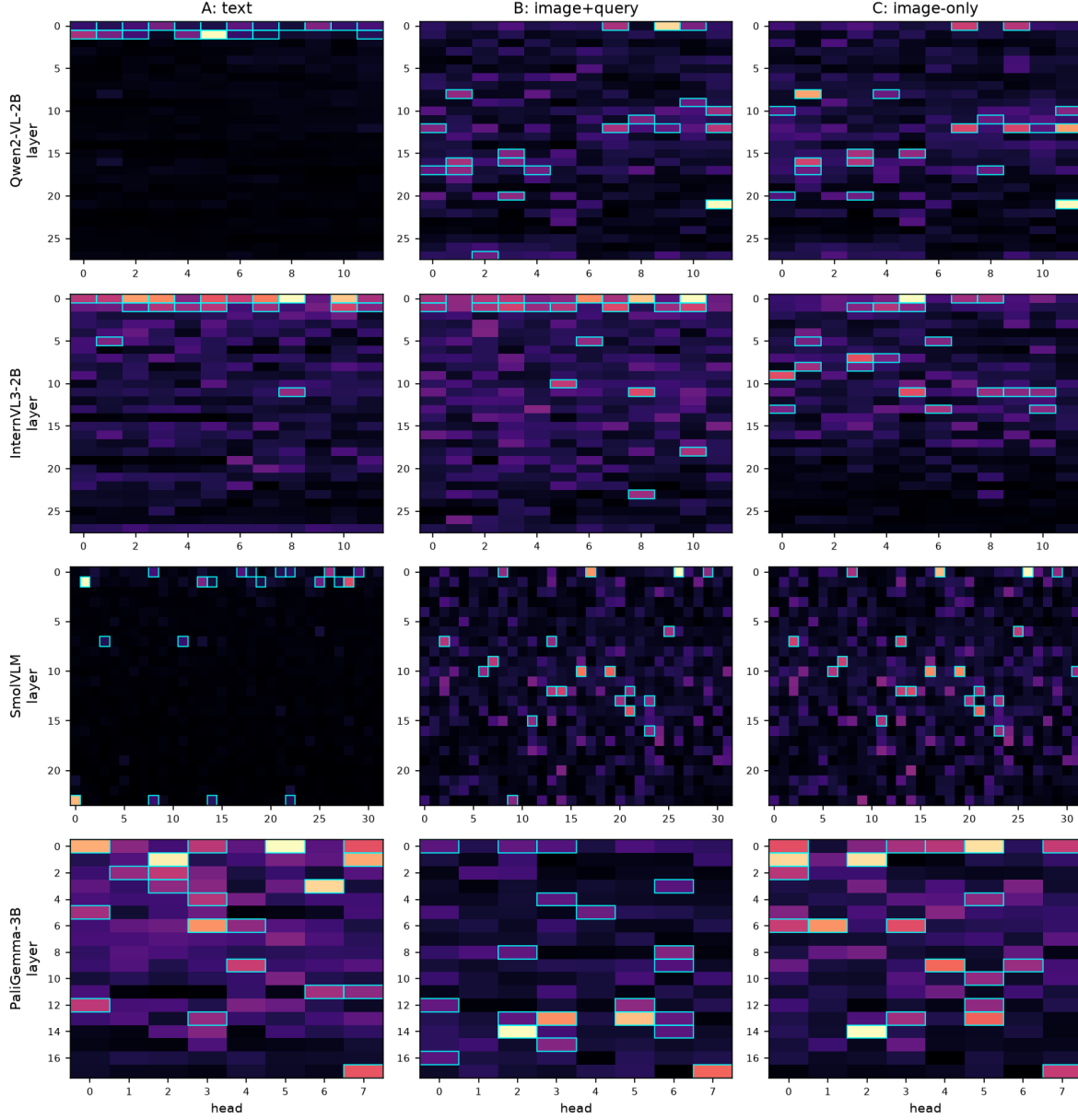


Figure 8: LAHIS per-head importance for Spanish, plotted as in Figure 7 (rows are the four models, columns the three routing conditions, cyan rings mark each panel’s top-20 heads). The important heads sit in the same layer bands as for Hindi, but for InternVL3-2B the text and image panels align less closely than they do for Hindi, consistent with its lower Spanish text-image overlap in Table 4.